

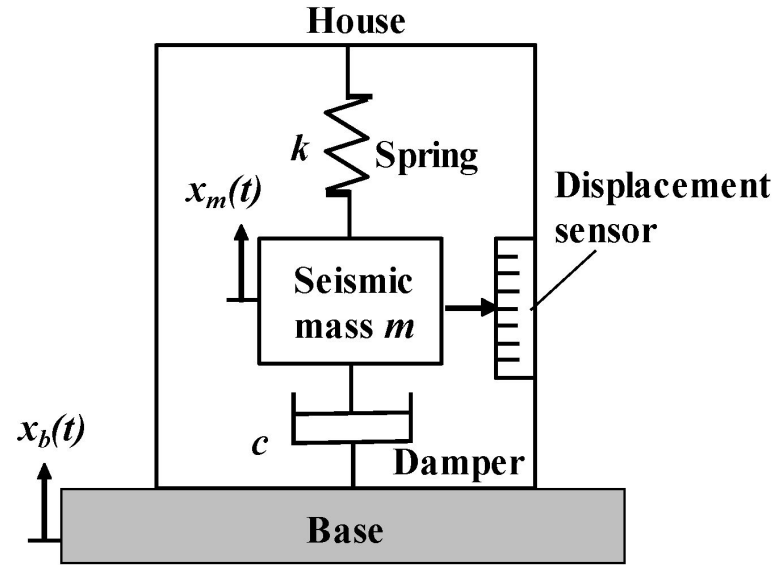
UAV attitude estimation

Group 1 - Balder, Svend and Christoffer

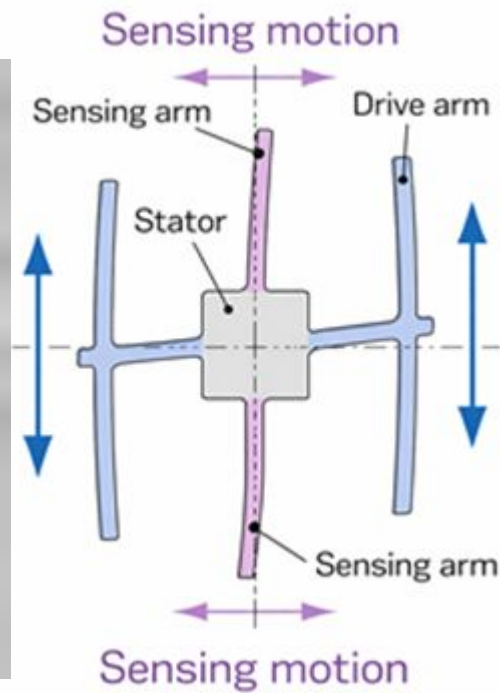
Disposition

- UAV attitude sensors
 - Accelerometer
 - Gyroscope
 - Compass
- UAV attitude sensing using accelerometers
 - Gimbal lock
- Gyro measurements
- Kalman filter

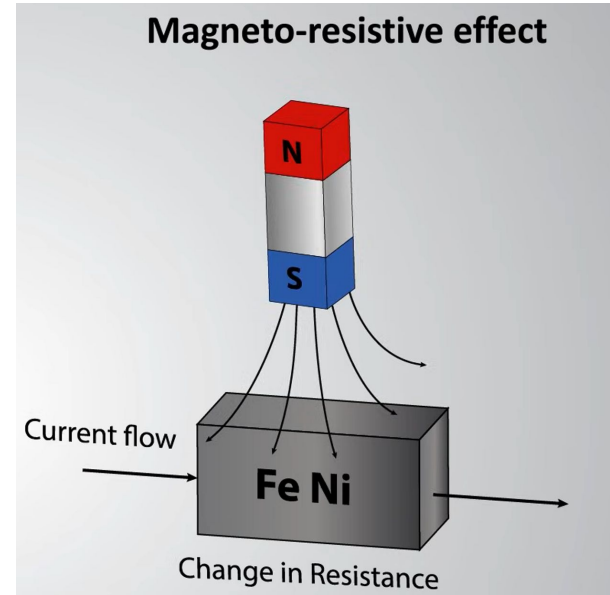
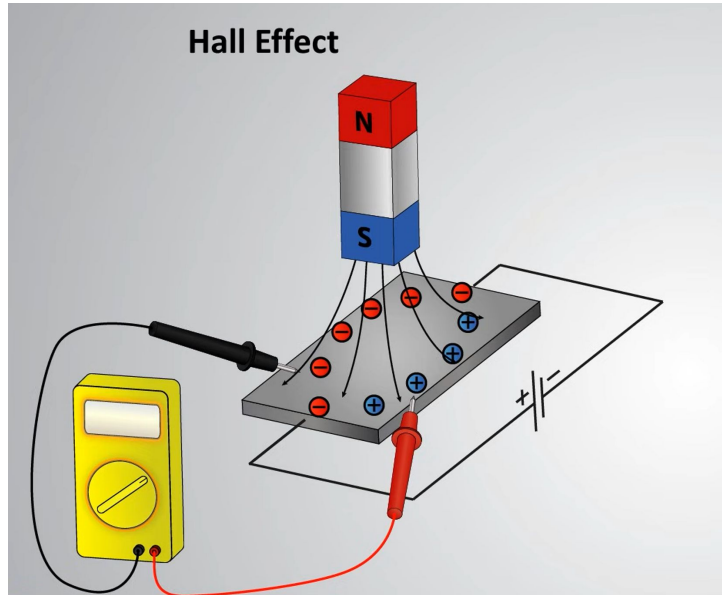
Accelerometer



Gyro



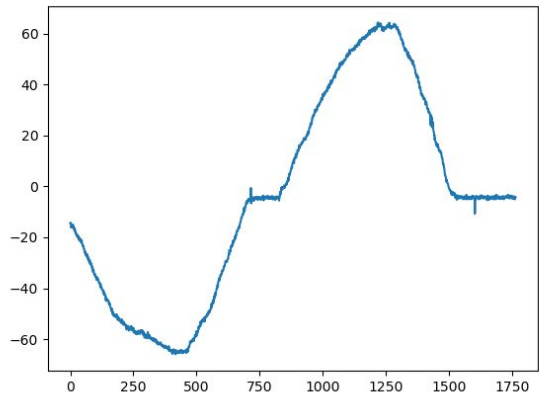
Magnetometer



3.2.1 & 3.2.2 UAV attitude sensing using accelerometers

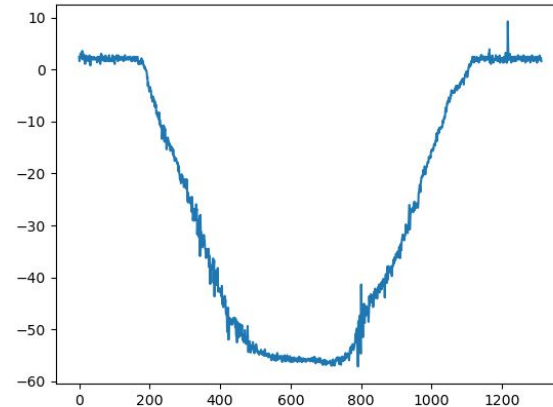
Roll 65 deg

$$\tan \theta_{yxz} = \left(\frac{-G_{px}}{G_{pz}} \right)$$

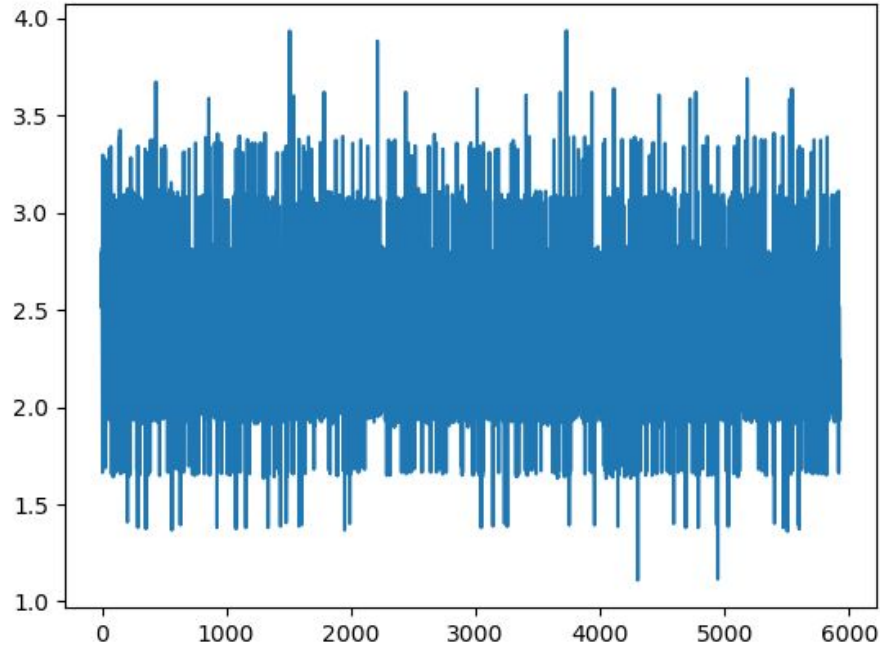


Pitch 55 deg

$$\tan \phi_{yxz} = \frac{G_{py}}{\sqrt{G_{px}^2 + G_{pz}^2}}$$



3.2.3 & 3.2.4 Accelerometer Noise

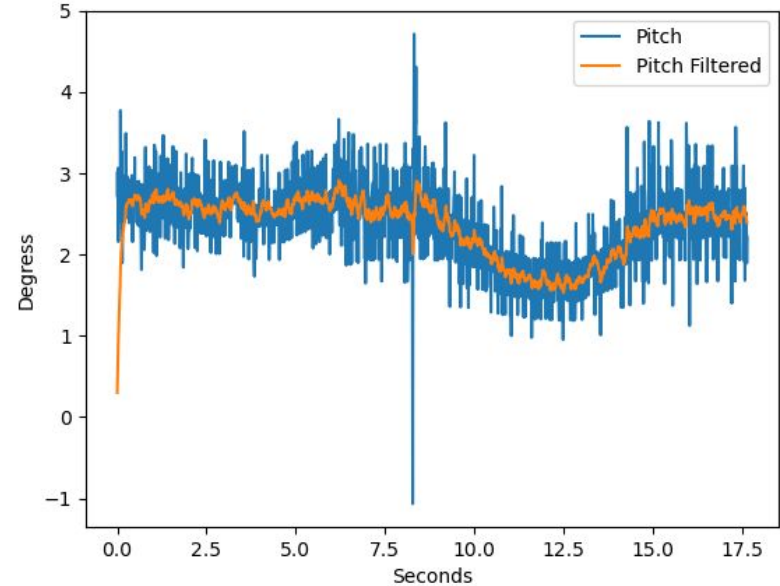
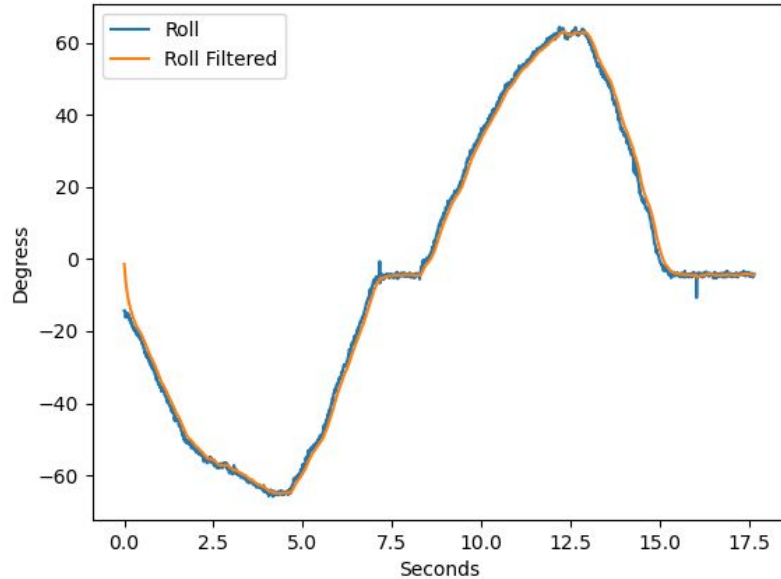


Noisy!!

Roll and pitch with lowpass filter

Simple filter:

$$x_{\text{filtered}} \leftarrow 0.90 \cdot x_{\text{filtered}} + 0.1 \cdot x_{\text{new}}$$



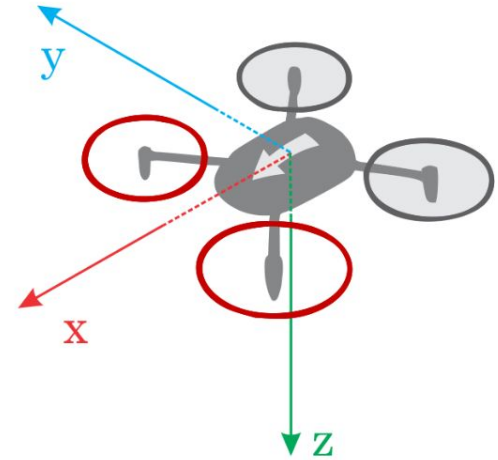
Gimbal Lock

Pitch

$$\tan \phi_{yxz} = \frac{G_{py}}{\sqrt{G_{px}^2 + G_{pz}^2}}$$

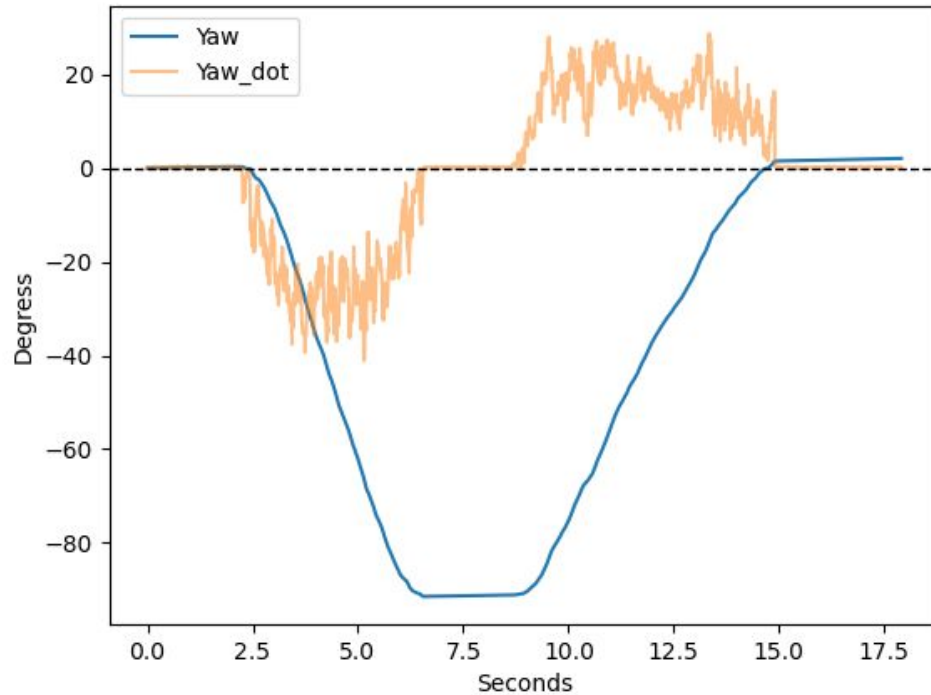
Roll

$$\tan \theta_{yxz} = \left(\frac{-G_{px}}{G_{pz}} \right)$$

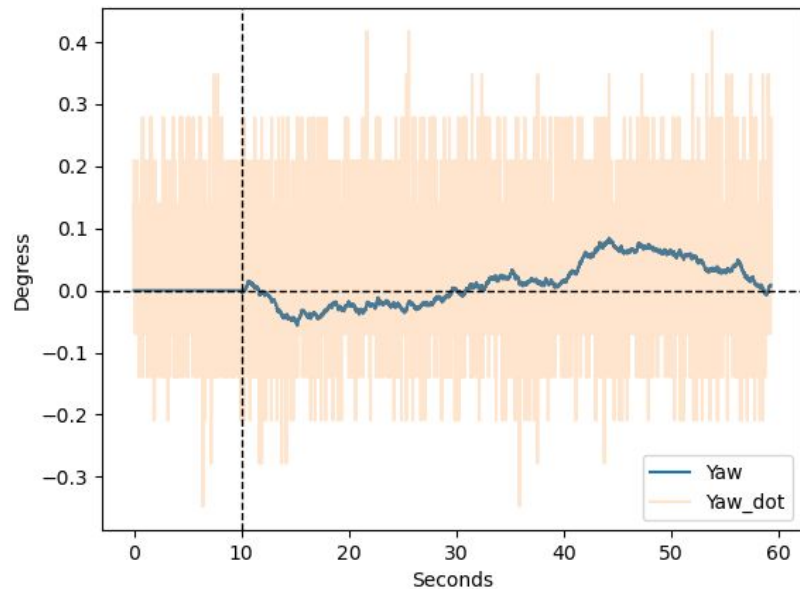
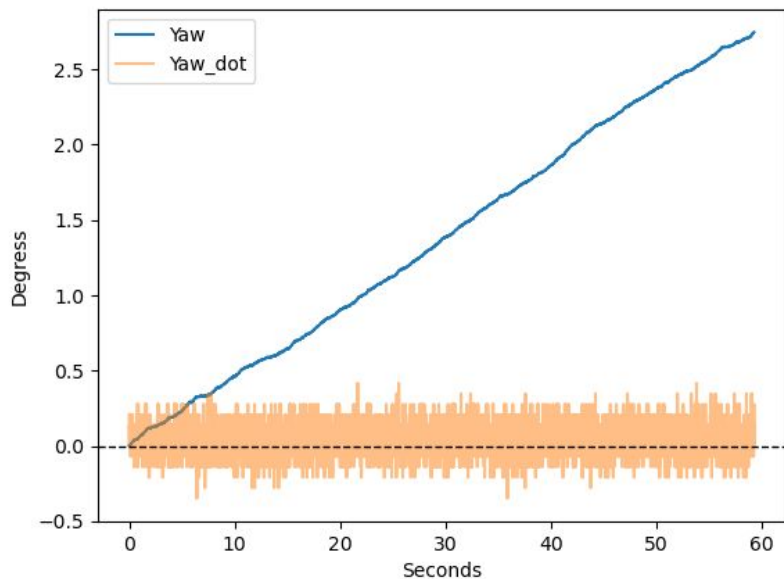


3.3.1 Gyro Measurements

90 degree turn



Gyro Bias (And Sources)



3.4.1 Kalman filter

